

Demonstrating High-Precision and High-Fidelity Digital Inking for Virtual Reality

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ABSTRACT

Pen computing has become popular with tablet and wall-screen computers for digital precision tasks such as writing, annotating, and drawing. Digital pens have been made possible by the developments in input sensing technologies integrated into such screens. Virtual Reality systems, however, largely detect input using cameras, whose update rates are insufficient for capturing pen input with the necessary fidelity. In this demonstration, we showcase a digital pen for VR that accurately digitizes writing and drawing, including small and quick turns. Our prototype Flashpen repurposes an optical flow sensor from gaming mice, which digitizes minute motions at over 8 kHz when dragged across a surface. We demonstrate several use-cases for Flashpen during interaction in VR, including sketching, selecting, annotating, and writing.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction devices—Graphics input devices;

1 INTRODUCTION

Alongside touch interaction, pen input has become a commodity modality on interactive devices, especially on tablets, but also on phones and wall-screen displays. With the growing availability of Virtual Reality platforms, researchers have also started to explore pen interaction in 3D spatial settings (e.g., for sketching, gestures input, or mid-air input techniques). To register input, current VR systems track the user's hands or handheld controllers through cameras built into the headset. Popular applications for sketching in mid-air include TiltBrush, GravitySketch, Quill and Logitech VR Ink Pilot. To bring actual pens into VR to replace hand-held controllers or finger painting, considerable research has been conducted to reduce the gap to analog interfaces, for example by using tablets to help draw in VR. In this demonstration, we show a prototype that further approaches analog pen input qualities in VR systems. Inspired by the inside-out tracking approach taken by the latest headsets, we present Flashpen, a stylus with an optical flow sensor at its tip. Flashpen's flow sensor is repurposed from high-end gaming mice and operates at a framerate of 8 kHz, thus digitizing pen strokes at a rate that is 40 times higher than that of state-of-the-art stylus input digitizers (e.g., Wacom, Microsoft Surface Pro).

2 FLASHPEN: ASSEMBLY AND IMPLEMENTATION [4]

We designed Flashpen to support fast interaction on any passive surface with the ambition to resolve pen strokes at the fidelity of analog handwriting. To that end, we designed a VR controller with the form factor of a pen. Flashpen's hinge allows the pen to adopt to comfortable postures during interaction on the surface so that users can write with it similar to a real pen but digitizing their ink in VR instead. In this section, we describe the components of our prototype

and their assembly followed by the VR and camera setup including our tracking pipeline.

2.1 Device Components

As shown in Figure 2 (right), we designed Flashpen to be held like an ordinary pen. All sensors and electronics used are depicted in Figure 1. The shell of the pen is made from a 30 mm plastic tube wrapped in black electrical tape. This tube encases all hardware components except for the optical flow sensor at the tip. When writing with a regular pen, users hold the device with different hand postures. To accommodate this with Flashpen, we used a light plastic joint (*LEGO Mindstorms* hinge, Figure 1.4) that we cut so as to be as close as possible to the surface. The hinge has an angular range of 110° from left to right. Flashpen can therefore be used by left-handed as well as right-handed people and supports various grips for holding.

To track relative motions, Flashpen embeds an optical flow sensor (Pixart PMW3360) at the tip of the pen (2) whose lens and cover (3) we sanded down to be as small as possible. With its flat tip, the design is similar to the Adonit Pro 3 [1]. An inertial sensor mounted inside the shell of the pen captures rotations and accelerations (InvenSense MPU6500). All parts connect to an Expressif ESP32 microcontroller that connects to the PC through a cable to minimize the latency of input sensing and for power. Reflective markers are mounted around the pen tip (1) for tracking absolute positions using cameras. Overall, Flashpen weighs approximately 200 g. We used the shown prototype for both the study and the video accompanying the paper.

2.2 Tracking Setup

The flow sensor at Flashpen's tip captures optical reflections at 8,000 fps and derives relative motions from inter-frame differences, represented through relative motions ($\Delta x, \Delta y$) encoded in fractions of a millimeter. We configured the flow sensor to operate in burst mode, such that Flashpen's microcontroller can report all relative motion events to the VR system at the full framerate. We complement the high-speed relative motion sensing on the tip with absolute positions from a camera system that records the reflections from Flashpen's

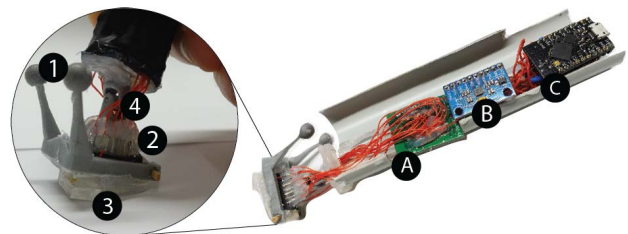


Figure 1: Prototype components: (1) markers for tracking, (2) optical flow sensor, (3) sensor lens, and (4) a plastic hinge to accommodate writing at various angles. Inside shell: (A) PMW3360 breakout board (optical flow sensor), (B) MPU6500 breakout board (IMU sensor), (C) ESP32 prototyping board (TinyPico)

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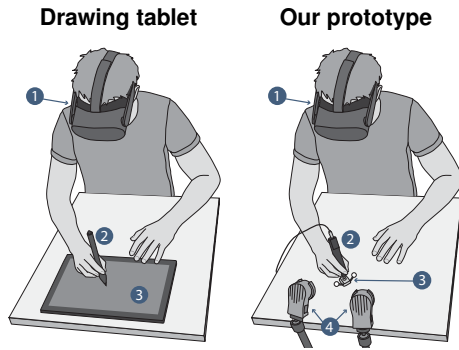


Figure 2: Virtual reality pen input based on a drawing tablet (left) versus our prototype (right). (1) Both variants use a VR headset. (2) Both, the drawing stylus (left) and our device (right) are held like a pen. (3) Conventional styli only work on the drawing tablet area (left), whereas we use an optical flow sensor that works on any surface (right). (4) We use two static OPTiTRACK FLEX13 cameras to retrieve the absolute pen position (when in mid-air) and the pen orientation.

markers. We perform basic blob detection and line intersection in the (pre-calibrated) stereo setup to resolve the pen's position and rotation in absolute 3D space.

3 APPLICATIONS

We designed a set of virtual reality application prototypes to showcase Flashpen's capabilities through VR interaction techniques. For all applications, Flashpen is represented as a virtual pen to maintain functional consistency between the physical world and the virtual world. The applications can be used on any surface orientation (i.e., horizontal or vertical).

Reviewing and annotating documents: In Figure 3A, a user browses a document library to select an article to review. The user highlights important parts, jots down comments, and emphasizes paragraphs of interest using Flashpen. Using a swipe gesture with the non-dominant hand (tracked through the VR headset), the user can switch between pages to review several pages while immersed in a VR environment. As a future use case, we envision that such an annotation application can also be mobile (e.g., on a notebook) to support on the go interaction while collaborating on documents. Related to this application, we see an opportunity for Flashpen as the basis for an exploration of broader scenarios and interaction techniques that could additionally involve touch in VR [3].

Multi-layered map: In Figure 3B, a user explores a multi-layered map with a magic lens metaphor [2]. The map has two layers, whereas the first layer is a geologic map and the second one is a satellite view seen through a magic lens. The user can use the pen as a precise manipulation tool to drag the magic lens around (Figure 3B, left), revealing the satellite layer. Instead of using a pre-defined round lens, the user can also precisely define a custom polygon shape (Figure 3B, right), e.g., following a road or a river.

Drawing application: We also demonstrate Flashpen's precision and mobility in a painting app (Figure 3C), which allows users to accurately fill shapes and interact with widgets. Examples for such widgets are the sliders that are displayed on the right. Users can put the pen down on a value to select or to drag the slider until reaching the desired color and thickness. Once a color has been selected, they can start drawing or filling monochrome shapes with color. Only with the jitter-free high precision of the flow sensor, small corners of the drawing can be accurately filled with color without error, which would be infeasible with cameras alone.

Motion path animation: As shown in Figure 3D, we use the high sample rate of Flashpen in this application to design motion expe-

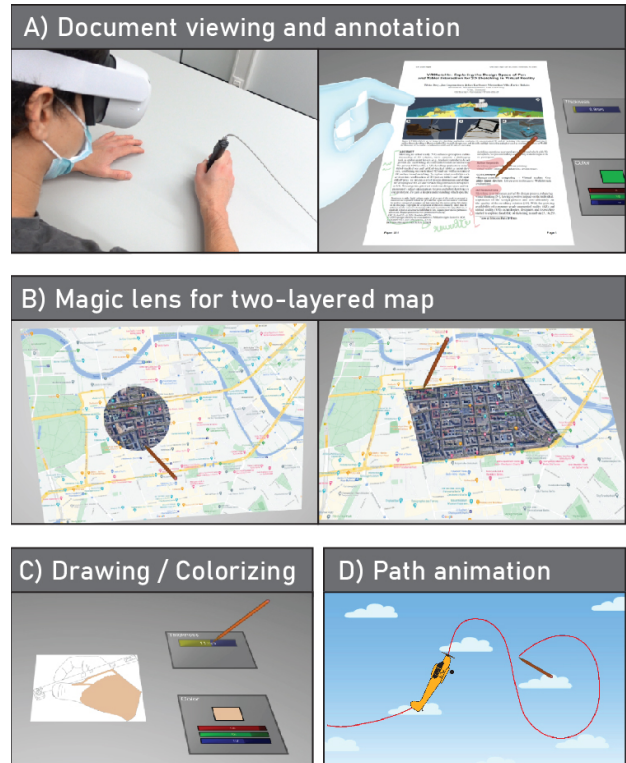


Figure 3: Virtual reality prototypes that utilize Flashpen. Each application can be used on a horizontal or vertical surface.

periences where the user draws strokes using varying input speeds to create a path animation of a plane. After drawing the spatiotemporal curve, the yellow plane will follow the exact same path while matching the user's velocity at every point on the curve. Thus, such motion paths simultaneously encode spatial as well as temporal information, e.g., by drawing slowly while going in a straight line and then accelerating while drawing a looping.

4 CONCLUSION

Our demonstration showcases high-fidelity pen input in Virtual Reality. We take a complementary approach of camera-based absolute tracking and pen-based high-speed inside-out tracking for fast-occurring relative motions. In a series of applications, we show the potential of inking in VR, whose quality and fidelity is uniquely enabled by our prototype Flashpen.

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